240481: porphyritic metadacite, Aphrodite prospect

(Black Flag Group, Kalgoorlie Terrane, Eastern Goldfields Superterrane, Yilgarn Craton)

Location and sampling

Kalgoorlie (SH 51-9), Bardoc (3137)
MGA Zone 51, 328991E 6659758N

Sampled on 8 August 2019

This sample was collected from the 420.20 – 422.83 m depth interval of drillcore 18APD012, drilled in 2018 by Bardoc Gold Limited at their Aphrodite prospect. The drillhole is located on Mount Vetters Station, about 5.0 km east of the Goldfields Highway, 3.9 km west of Upper Scotia Dam, and 1.7 km north of Middle Scotia Dam.

Geological context

The unit sampled is a metadacite of the 2692–2655 Ma Black Flag Group of the Kalgoorlie Terrane (Witt, 1994; Smithies et al., 2018; Hall et al., 2019). The Black Flag Group consists mainly of felsic metavolcaniclastic and metasedimentary rocks with subordinate felsic and mafic volcanic rocks. The Black Flag Group overlies the 2753–2680 Ma Kalgoorlie Group, is unconformably overlain by the 2664–2640 Ma Merougil Formation (Hall et al., 2019), and in part is intruded by the c. 2680 Ma Golden Mile Dolerite (Rasmussen et al., 2009). The sample is from the lower part of the Black Flag Group, in the eastern limb of the Goongarrie – Mount Pleasant Anticline near the contact with the underlying Victorious Basalt of the Vetters Subgroup of the Kalgoorlie Group (Witt, 1994). The metadacite is underlain by a 3 m-thick unit of black shale that conformably overlies the plagioclase-phyric Victorious Basalt. The sample was collected to verify its assignment to the Black Flag Group. A granodiorite about 15.8 km to the southeast yielded an igneous crystallization age of 2657 ± 5 Ma (GSWA 93901, Nelson, 1995). An igneous crystallization age of 2661 ± 3 Ma was reported for a granophyric dolerite dyke, sampled about 16.1 km to the south-southeast (GSWA 142858, Nelson, 1998).

Petrographic description

The sample is a medium-grained, plagioclase-phyric metadacite (Fig. 1), consisting of about 50–55% groundmass, 30% plagioclase, 15% biotite-replaced phenocrysts, 1–2% sulfide minerals, and accessory apatite, titanite and zircon. The groundmass is massive but preserves a finely micropoikilitic texture that indicates it was originally glassy. It is composed of microcrystalline, poorly defined laths of abundant probable albite and less abundant interstitial quartz. Plagioclase phenocrysts (andesine, $An_{34–38}$) are subhedral to euhedral, equant to tabular, up to 2.5 mm in size, randomly oriented, and variably altered to sericite and trace carbonate. The biotite-replaced phenocrysts are up to 2 mm in size and many exhibit a rhombic shape typical of amphibole, indicating that most were originally hornblende. A small proportion are tabular, up to 2 mm long, and exhibit a lamellar texture, suggesting they may have been biotite phenocrysts. The replacement biotite is pale tan to orange/brown and occurs as decussate aggregates of poorly defined anhedral laths, mostly less than 0.3 mm long. Biotite in the replaced phenocrysts is slightly altered to chlorite and these aggregates also include minor carbonate and sulfide minerals and trace titanite up to 0.1 mm in size. Sulfide minerals consist of highly irregular grains up to 0.5 mm in size of pyrite, less abundant pyrrhotite, and trace chalcopyrite, all mainly occurring within the biotite-replaced phenocrysts. Apatite is euhedral and bladed and up to 0.3 mm in size. Anhedral pink titanite is up to 0.2 mm in size. The rock is not obviously deformed and the decussate texture of the biotite that has replaced original hornblende or biotite phenocrysts may reflect contact metamorphism at biotite grade.

Zircon morphology

Zircons isolated from this sample are colourless and mainly euhedral. The crystals are up to 150 µm long, and elongate, with aspect ratios up to 4:1. In cathodoluminescence (CL) images, concentric zoning is ubiquitous. A CL image of all zircons is shown in Figure 2.

Analytical details

This sample was analysed on 25–26 February 2021, using SHRIMP-B. Twelve analyses of the M257 standard were obtained during the session, of which 11 analyses indicated an external spot-to-spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.18% (1σ). Isotopic mass fractionation of $^{207}Pb/^{206}Pb$ ratios during the session was corrected by reference to the OGC1 standard; measured ratios were increased by 0.22%. Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ ratios and dates listed in Table 1. Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Twenty-two analyses were obtained from 22 zircons. Results are listed in Table 1, and shown in concordia diagrams (Figs 3, 4).

Figure 1. **Drillcore image for sample 240481: porphyritic metadacite, Aphrodite prospect**

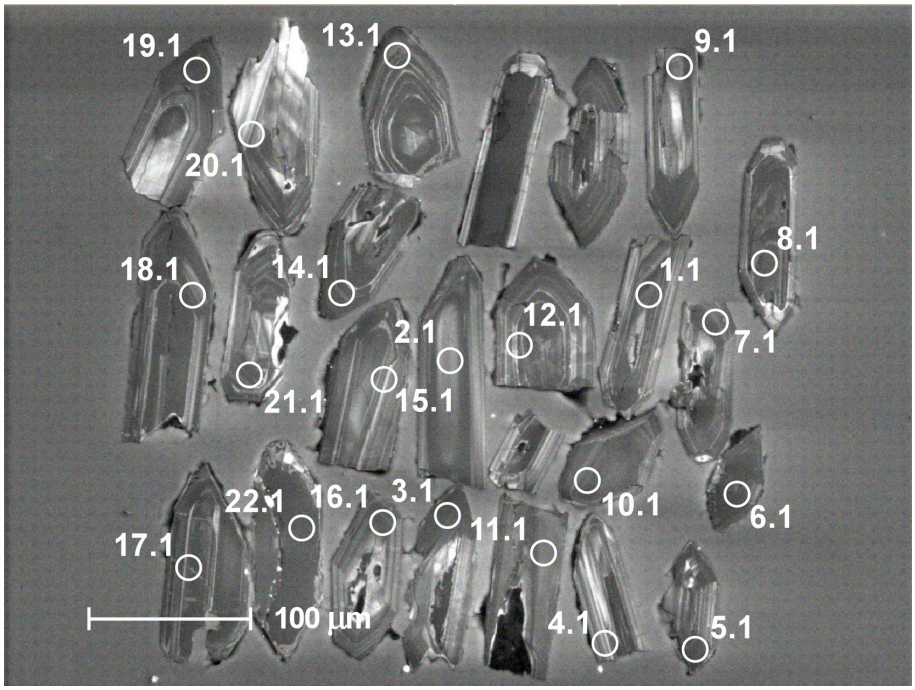

Figure 2. **Cathodoluminescence image of representative zircons from sample 240481: porphyritic metadacite, Aphrodite prospect. Numbered circles indicate the approximate locations of analysis sites**

Table 1. Ion microprobe analytical results for zircons from sample 240481: porphyritic metadacite, Aphrodite prospect

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	$f204$ (%)	$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		$^{238}U/^{206}Pb^*$ ± 1σ		$^{207}Pb^*/^{206}Pb^*$ ± 1σ		$^{238}U/^{206}Pb^*$ date (Ma) ±1σ		$^{207}Pb^*/^{206}Pb^*$ date (Ma) ±1σ		Disc. (%)
I	20	20.1	85	47	0.55	0.292	2.061	0.026	0.18474	0.00142	2.067	0.026	0.18213	0.00161	2543	26	2672	15	4.8
I	5	5.1	362	359	0.99	0.035	2.007	0.016	0.18290	0.00082	2.008	0.016	0.18259	0.00083	2605	18	2677	8	2.7
I	1	1.1	198	165	0.83	0.033	1.980	0.025	0.18291	0.00185	1.981	0.025	0.18262	0.00185	2635	27	2677	17	1.6
I	21	21.1	355	257	0.73	0.040	1.982	0.016	0.18304	0.00086	1.983	0.016	0.18268	0.00087	2633	18	2677	8	1.7
I	9	9.1	175	155	0.89	0.086	2.007	0.019	0.18367	0.00102	2.009	0.019	0.18290	0.00105	2604	20	2679	10	2.8
I	6	6.1	290	305	1.05	0.076	1.991	0.016	0.18391	0.00085	1.992	0.016	0.18323	0.00087	2622	18	2682	8	2.2
I	16	16.1	269	276	1.02	0.128	1.971	0.018	0.18458	0.00096	1.974	0.018	0.18343	0.00101	2642	20	2684	9	1.6
I	18	18.1	271	291	1.07	0.014	2.011	0.017	0.18419	0.00139	2.011	0.017	0.18406	0.00139	2602	19	2690	12	3.3
I	3	3.1	352	350	0.99	0.010	1.920	0.015	0.18427	0.00079	1.920	0.015	0.18418	0.00079	2702	18	2691	7	-0.4
I	22	22.1	450	475	1.05	-0.012	1.922	0.015	0.18439	0.00120	1.921	0.015	0.18450	0.00120	2701	18	2694	11	-0.3
I	14	14.1	258	242	0.94	0.028	2.050	0.017	0.18475	0.00088	2.051	0.017	0.18451	0.00089	2561	18	2694	8	4.9
I	7	7.1	266	242	0.91	0.028	1.974	0.017	0.18478	0.00088	1.975	0.017	0.18453	0.00089	2641	19	2694	8	2.0
I	12	12.1	236	244	1.03	-0.023	1.970	0.017	0.18437	0.00091	1.969	0.017	0.18458	0.00092	2647	19	2694	8	1.8
I	17	17.1	452	542	1.20	0.045	1.969	0.022	0.18504	0.00079	1.970	0.022	0.18464	0.00080	2647	24	2695	7	1.8
I	10	10.1	252	196	0.78	0.066	1.933	0.017	0.18554	0.00093	1.934	0.017	0.18494	0.00095	2687	20	2698	8	0.4
I	19	19.1	304	300	0.99	0.019	2.012	0.017	0.18531	0.00085	2.012	0.017	0.18515	0.00085	2601	18	2700	8	3.7
I	2	2.1	238	287	1.21	0.039	2.011	0.018	0.18561	0.00101	2.012	0.018	0.18525	0.00102	2601	19	2700	9	3.7
I	15	15.1	197	147	0.75	0.045	1.940	0.019	0.18571	0.00104	1.940	0.019	0.18530	0.00106	2680	22	2701	9	0.8
X	11	11.1	555	892	1.61	0.012	1.820	0.014	0.18951	0.00075	1.820	0.014	0.18940	0.00075	2823	18	2737	7	-3.2
D	8	8.1	916	799	0.87	0.053	2.684	0.029	0.14820	0.00087	2.686	0.029	0.14773	0.00088	2040	19	2320	10	12.0
D	4	4.1	267	432	1.61	0.000	2.459	0.022	0.15298	0.00084	2.459	0.022	0.15298	0.00084	2200	17	2380	9	7.5
D	13	13.1	362	338	0.93	0.014	2.189	0.017	0.17627	0.00076	2.190	0.017	0.17615	0.00076	2425	16	2617	7	7.3

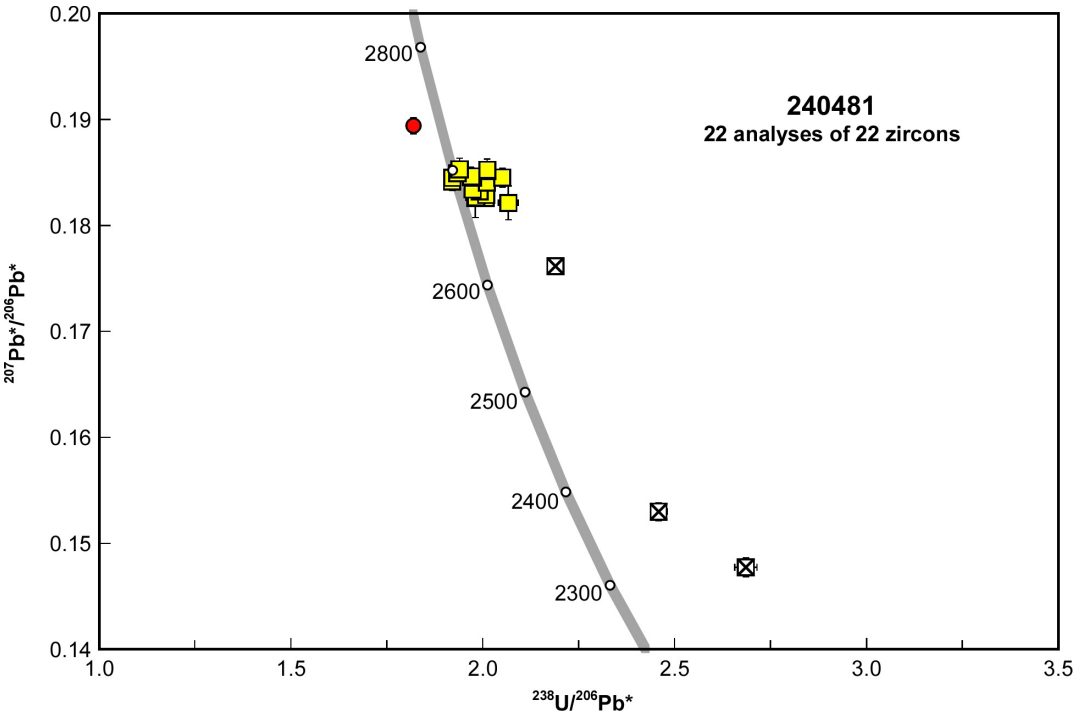

Figure 3. U–Pb analytical data for sample 240481: porphyritic metadacite, Aphrodite prospect. Yellow squares indicate Group I (magmatic zircons); red circle indicates Group X (xenocrystic zircon); crossed squares indicate Group D (discordance >5%)

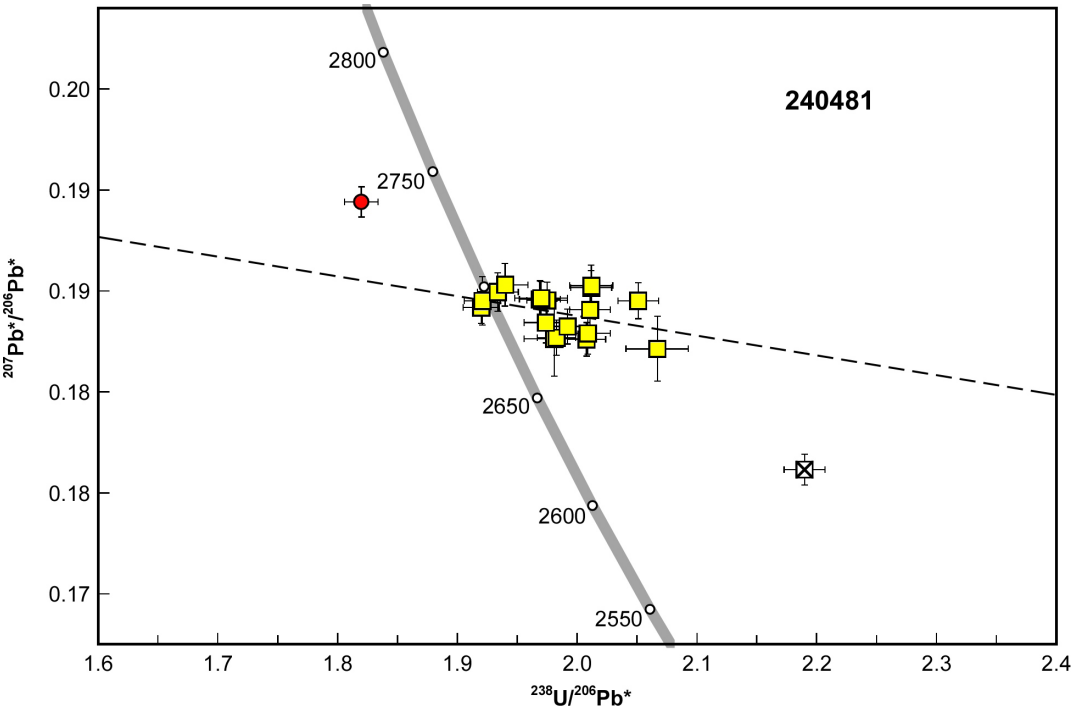

Figure 4. Expanded view of U–Pb analytical data for sample 240481: porphyritic metadacite, Aphrodite prospect. The dashed line indicates a regression through data in Group I. Symbols as in Figure 3

Interpretation

The analyses are concordant to moderately discordant (Fig. 3). Three analyses are >5% discordant. The dates obtained from these three analyses (Group D; Table 1) are unreliable, and are considered not to be geologically significant. The remaining 19 analyses can be divided into two groups, based on their $^{207}Pb^*/^{206}Pb^*$ ratios.

Group I comprises 18 analyses (Table 1), which yield a discordia with intercepts at 2694 ± 8 and c. 400 Ma (MSWD = 0.96).

Group X comprises one analysis (Table 1), which yields a $^{207}Pb^*/^{206}Pb^*$ date of 2737 ± 7 Ma (1σ).

The date of 2694 ± 8 Ma for the 18 analyses in Group I is interpreted as the age of igneous crystallization. The date of 2737 ± 7 Ma (1σ) for the single analysis in Group X is interpreted as the age of an inherited zircon.

References

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Recommended reference for this publication

Wingate, MTD, Lu, Y, Fielding, IOH and De Paoli, M 2022, 240481: porphyritic metadacite, Aphrodite prospect; Geochronology Record 1859: Geological Survey of Western Australia, 5p.

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Data released: 29 April 2022